Races Decided by the Born Rule

The complex instance of graded where-clauses: a quantum namespace, contention-set completion, and the fragment on which a process term denotes a quantum state

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Abstract

The rho calculus does not say how the non-determinism of a race is resolved. An implementation must supply that mechanism, and quantum mechanics is one of the available choices: it supplies amplitudes for the branches of a race and the Born rule for turning them into outcomes. This note takes that instantiation seriously and asks what has to be true of a program before its denotation is a quantum state.

The design is one line of grammar, carried over from the companion paper on resolution algebras: the `where` clause of a for-comprehension is valued in a commutative semiring rather than in the Booleans, and is evaluated per candidate match. Here $\mathbb{V} = \mathbb{C}$ throughout.

Four things are new. First, a *quantum namespace*: names bound by `new` are unforgeable and are the only ones on which a clause may take complex values, while quoted names are classical. Unique ownership of quantum data — which existing quantum process calculi obtain from a type system — follows from unforgeability, and the herald of a non-unitary step lands in the classical namespace by construction, which is what gives measurement a place to happen. Second, the denotation is *register-indexed*: $[\![P]\!]$ lives in a Fock space over the names P can act on, and parallel composition combines states by adding occupancies. Because parallel composition is associative-commutative, the multi-occupancy sectors are symmetric, so bosonic statistics and the permanent of the scattering matrix are forced by structural congruence rather than chosen. Third, the completion condition that makes a step norm-preserving is stated over the *contention set* — the connected components of the conflict relation, with maximal matchings as the alternatives and the fibre-summed amplitudes normalised — and conflict components tensor rather than sum. The race equation, the join, and the two-receipt race are the special cases. Fourth, we identify the fragment on which the one-step operator is an isometry. It is not the whole calculus: absorbing normal forms and spectator tokens each break it, and both are visible syntactically. Under halt tagging, staging, and a dual-rail encoding the failures disappear in every case we have been able to construct.

The dual-rail encoding also settles a question the previous version got wrong. Under the payload encoding, a step that uses each bound name exactly once is block-diagonal in the input basis, hence not a gate; that is a fact about the encoding, and under dual rail every 2×2 and the entangling 4×4s are realised with no discard anywhere. Shor's algorithm is rebuilt on that basis, with the whole term given, generated from the same gate list the simulator runs.

Non-unitary clauses are contained rather than excluded: contractivity is a judgement, the defect is dilated at the site, and the ancilla is neither traced nor projected but written as a classical message which a retry loop may read and the resource ledger charges. What was free post-selection becomes a bill of 2^n firings.

We are explicit about what is not established: norm preservation is proved at a configuration and conjectured for arbitrary concurrency; the staging condition is a measured

survey and not a theorem; the complexity containment is a conjecture; and quantum alternation is known to be incompatible with the standard fixed-point semantics for recursion, which our budget does not repair.

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1 Introduction

1.1 Where the choice is

A rho program does not determine its own execution. Write

$$x!(Q_1) \mid \mathsf{for}(y \leftarrow x)P \mid x!(Q_2) \tag{1}$$

and the operational semantics permits two reductions. It does not say which one happens, and it does not say with what tendency. This is not an oversight; it is the defining feature of a mobile process calculus, and it is what makes the calculus a specification of interaction rather than of a machine.

An implementation must close the gap. RSpace closes it by presenting the parallel fragment as a store from channels to polarity-homogeneous bags and letting the physical race between cores decide which produce is zipped with which consume. That is a resolution mechanism, and it is a choice.

The resolution mechanism is a *parameter*, and other values of the parameter are available. Quantum mechanics is one: it supplies amplitudes for the branches of a race and the Born rule for turning them into outcomes. This note develops that instantiation. The companion paper [24] develops the parameter itself — the resolution algebra, the conservativity of the Boolean value, the normal-form theorem for weight maps — and a second companion [21] develops the non-negative real instance and its use in inference-driven execution. Everything here is $\mathbb{V} = \mathbb{C}$.

1.2 What this paper must answer

The previous version of this material [13] was reviewed, and the review was right about four things. Non-contractive clauses were legal but had no dilation. Dilating a receipt site does not make the one-step operator norm-preserving, because the operator sums over sites as well as over candidates. The herald was simultaneously a coherent ancilla and an ordinary message, with no account of the transition between them. And the stage-one proposition had a counterexample.

Two further defects were not raised and are addressed here. Norm preservation of each column is only half of $T^\dagger T = \mathbb{1}$; the other half is that distinct columns be orthogonal, and that fails for reasons having nothing to do with weights. And a process term is not, in general, the carrier of a quantum state at all — states must be indexed by a register, and states over overlapping registers do not combine.

1.3 Contributions

1. **The quantum namespace** (§6). Names bound by `new` are unforgeable; quoted names are not. Take the first as the quantum namespace. Unique ownership follows from unforgeability rather than from a type discipline, and the herald channel $h(r) := @r$ is classical because it is a quotation — so the classical/quantum boundary, which is where measurement happens, is drawn by the calculus and not by us.

2. **The denotation is register-indexed** (§7). $[\![P]\!]$ lives in a Fock space over the names P can act on. Composition adds occupancies, and is injective exactly when the registers are disjoint; overlapping registers forget which side supplied which quanta, which is indistinguishability. Since $\mid$ is associative-commutative the sectors are symmetric, so **bosonic statistics are forced by structural congruence** (Proposition 7.2).

3. **Contention-set completion** (§8). The unit of normalisation is the connected component of the conflict relation; the alternatives within a component are its maximal matchings; the fibre-summed amplitudes are what must have unit norm; and components tensor rather

than sum. Proposition 8.2 and Table 1, which measures the three wrong answers against the right one.

4. **The quantum-safe fragment** (§10). Column orthogonality fails in two ways, both syntactic: *absorption*, where a normal form idles into a configuration another branch is still flowing towards, and *spectators*, where a parked token is indistinguishable from an emitted one. Halt tagging removes the first, staging removes the second, and dual rail removes both. Table 2.
5. **Linearity does not cost universality** (§11). The previous version proved that a step using every bound name exactly once is block-diagonal in the input basis. The proposition is true and its hypothesis is a property of the encoding. Under dual rail every 2×2 is realised with no discard (Lemma 11.4), and so is every entangling 4×4 (Lemma 11.5).
6. **Shor, in full** (§12). The whole term, gadget contracts and instance, generated from the gate list the simulator runs.
7. **Heralds and the price** (§13). Contractivity is a judgement; the defect is dilated at the site; the ancilla is written as a classical message; post-selection becomes a retry loop that the ledger charges at 2^n firings.
8. **Position** (§14). The right comparison class is quantum control, not quantum process calculus. Two of Bădescu and Panangaden's negative results [4] are discharged by the design and one is not; their suggested way out of the quantum-alternation impasse — interferometry with no distinct control qubit — is this design.

1.4 Reading order

§2 fixes the language, including the third receipt arrow $\multimapinv$ that the linear-binding discipline needs. §§3–5 are background and are close to the previous version. §§6–10 are the new semantics and can be read on their own. §§11–12 are the worked example, and §13 the containment.

1.5 What this paper does not do

It does not prove a quantum advantage. It does not prove Conjecture 13.11. It proves norm preservation at a configuration and conjectures it for arbitrary concurrency (Conjecture 8.6). The staging condition of §10 is a measured survey over small systems, not a theorem. And it does not solve the recursion problem of Obligation 15.3.

2 The calculus

This section fixes the language. It is the rho calculus [18] with three changes: the `where` clause of a for-comprehension takes values in a commutative semiring rather than in the Booleans, the clause is evaluated once per candidate match rather than once per communication, and a third receipt arrow is added for the binding discipline of §11.

2.1 Terms and names

Definition 2.1 (Syntax)**.** Processes and names are given mutually:

$$
\begin{array}{rcll}
P, Q & ::= & 0 & \text{the stopped process} \\
& \mid & x!(P) & \text{output} \\
& \mid & \mathsf{for}(\, y_1 \,\alpha\, x_1 \;\&\; \cdots \;\&\; y_m \,\alpha\, x_m \;\mathsf{where}\; \psi\,)P & \text{input, } m \geq 1 \\
& \mid & P \mid Q & \text{parallel composition} \\
& \mid & *x & \text{drop} \\
& \mid & \mathsf{new}\; x \;\mathsf{in}\; P & \text{fresh name} \\
x, y & ::= & @P & \text{quote} \\
& \mid & \text{a name bound by } \mathsf{new} \text{ or by an input prefix} & \\
\alpha & ::= & \Leftarrow \mid \leftarrow \mid \multimapinv & \text{the receipt arrows, below}
\end{array}
$$

A term with $m > 1$ patterns is a *join*; the patterns need not sit on distinct channels, and when they do not the receipt is contended with itself.

Names are quoted processes and $*x$ drops a name back to the process it quotes. This is the whole of reflection, and it is why the calculus needs no separate theory of higher-order values: a channel is a program and a program is a channel. It is also the feature that makes Lemma 11.4 possible, since a name received at runtime may be used as the channel of a send.

2.2 Structural congruence

Definition 2.2 (Congruence)**.** $\equiv$ is the least congruence such that $(\mathcal{P}, |, 0)$ is a commutative monoid, together with $*@P \equiv P$ and $@*x = x$, closed under α-equivalence and under scope extrusion for new.

Commutativity of $|$ is not a convenience. A configuration therefore records *how many* messages rest on a channel and not which is which, which is what Proposition 7.2 turns into a statement about statistics.

2.3 The three receipt arrows

Definition 2.3 (Receipts)**.** Write

$$\mathsf{for}(y \Leftarrow x)P, \qquad \mathsf{for}(y \leftarrow x)P, \qquad \mathsf{for}(y \multimapinv x)P$$

for, respectively: a *persistent* receipt, which survives its own firing; a *linear* receipt, which is consumed by it; and a linear receipt whose bound names are additionally used *exactly once* in the continuation (Definition 11.1). In program text the third is written `o-`, the reverse of linear logic's lollipop [11]: the head sits at the bound name, as it does in $\leftarrow$, because the datum flows from the channel to the variable.

Remark 2.4 (Receipts are linear by default). Linearity of the *receipt* is what makes the surviving message of (1) a which-path record, and hence what Proposition 4.1 rests on: under a persistent receipt both messages are eventually consumed, the receipt survives either way, and the two derivations reconverge with nothing left to weight. At a join persistence is harmless but idle, since a join consumes all the data at its channel and the surviving receipt is identical across matchings, so it distinguishes none of them. Persistence and linear binding are independent: a persistent receipt consumes each datum once and uses its bound names once per firing, which is why the installer contracts of §12 are legal.

The three arrows are three disciplines, not three sorts of channel. Whether a channel is quantum is settled by Definition 6.1, and $\multimapinv$ is well formed only there.

2.4 Reduction

Definition 2.5 (Candidates)**.** At a receipt r in a term P, a *candidate* is an injective assignment σ of r's patterns to messages resting on the corresponding channels. $\mathrm{Cand}(P, r)$ is the set of candidates.

Definition 2.6 (Communication)**.** For a candidate σ of a receipt r with clause ψ and continuation P,

$$\mathsf{for}(\,\vec{y}\,\alpha\,\vec{x}\ \mathsf{where}\ \psi\,)P \ \mid\ x_1!(Q_1) \mid \cdots \mid x_m!(Q_m) \xrightarrow{[\![\psi]\!](\sigma)} P\{@Q_{\sigma(1)}/y_1, \ldots, @Q_{\sigma(m)}/y_m\},$$

the receipt being discarded for $\leftarrow$ and $\multimapinv$ and retained for $\Leftarrow$. The transition carries the clause's value at σ as its weight. A candidate of weight 0 does not fire, so the crisp sort of the clause keeps its gating job and the graded sort does nothing else. Reduction is closed under $|$, under new, and under $\equiv$.

Definition 2.7 (Well-formedness)**.** A term is *well formed* when every receipt using $\multimapinv$ is on quantum channels (Definition 6.1) and satisfies Definition 11.1; every clause taking values outside $\mathbb{B}$ is at a receipt on quantum channels; and every such clause is contractive (Definition 13.2). Nothing else is refused.

A single rule with a weight is not yet a semantics, because a term generally offers many candidates at many receipts and the rule says nothing about how their weights relate. Supplying that relation is the work of §8, and getting it wrong at the level of the receipt rather than the contention set is the defect this version repairs.

2.5 What is not primitive

There is no replication and no summation. Recursion is written with reflection, by a receipt whose continuation re-installs a quotation of itself; this is why Obligation 15.3 is about reflection rather than about a fixed-point combinator. There is no restriction operator in the original presentation either: freshness is obtained by quoting an unguessable process. We keep new as a primitive binder because §6 makes its binding structure carry the classical/quantum distinction, and a derived form would leave that distinction without a syntactic anchor.

3 Why context-based full abstraction cannot help

Going from the pi calculus into rho, the question of how non-determinism is resolved never arises, because both languages are silent in the same way. Mapping rho to quantum mechanics is different: quantum mechanics makes a definite commitment about resolution, namely the Born rule. A repair that enlarges the class of rho contexts therefore cannot expose the additional structure, because contexts can *create* apertures — add contention — but cannot *read* them. Every context is itself resolved by the same unspecified mechanism.

This is not merely an observation about rho.

Proposition 3.1 (No extensional semantics, after [4])**.** *A conditional that applies a global phase in one branch and the identity in the other implements a controlled phase, although the two branches are physically indistinguishable as quantum operations. Hence there is no structural semantics for quantum alternation based on operations with extensional equality.*

Our situation is the process-calculus shadow of that statement: the resolver is exactly the datum that an extensional semantics discards, so no amount of context can recover it.

Definition 3.2 (Resolver separation)**.** Fix a family $\mathcal{R}$ of resolvers. The object of study is $\mathcal{R} \times \mathcal{P} \to$ distributions over normal forms. Two resolvers are *separated* at P when their images at P differ. Quantum advantage, if it exists, is a separation, not a full-abstraction theorem.

Definition 3.3 (Support adequacy and the interference deficit)**.** The one-sided obligation on a resolver is $\mathrm{supp}(\llbracket P \rrbracket) \subseteq \mathrm{Reach}(P)$. The *interference deficit* $\mathcal{D}(P) := \mathrm{Reach}(P) \setminus \mathrm{supp}(\llbracket P \rrbracket)$ is the set of operationally reachable normal forms carrying amplitude zero. Destructive interference is exactly the failure of the reverse inclusion.

Contexts remain useful, but as apparatus rather than as observers: an arrangement that converts a phase difference into a difference of support is an interferometer, and §9 builds the smallest one.

4 What unmodified rho already gives

Read (1) as an equational constraint with uniform weights. The question is which derivations can reconverge, since interference requires reconvergence and a non-zero relative phase and nothing else.

The previous version asserted that reconvergence in unmodified rho requires structurally congruent racing messages. That is false, and the counterexample is instructive. Take $Q_1 \not\equiv Q_2$ and

$$x!(Q_1) \mid x!(Q_2) \mid \mathsf{for}(y \leftarrow x)\{a!(0)\} \mid \mathsf{for}(z \leftarrow x)\{b!(0)\}.$$

The two complete matchings send Q_1 to y and Q_2 to z, or the reverse. Both yield $a!(0) \mid b!(0)$. Non-congruent racers reconverge.

They reconverge because y and z are *discarded.* That is the general mechanism, and it is worth stating in the form that survives.

Proposition 4.1 (Stage one, restated)**.** *Under* (1) *with uniform non-negative weights, two derivations reconverge exactly when the outcome does not record which datum went where. Congruence of the racing messages is one way to achieve this; discarding the bound names is another. In either case the interference is constructive, since non-negative reals cannot cancel.*

The example is not an anomaly but an instance: it is a *distributed join*, the case of Theorem 9.2 in which the patterns of a contention set are spread over several receipts rather than gathered into one. The constructive half of the old proposition survives; the confinement to congruent racers does not, and its loss is a gain — unmodified rho interferes more often than we had claimed, and always for the same reason.

5 The graded clause, in brief

The syntax is fixed in §2; the algebra is developed in [24]. This section records what is needed here.

The `where` clause is two-sorted. The *crisp* sort β is the existing condition language, with the full Boolean structure. The *graded* sort takes values in a commutative semiring $\mathbb{V}$:

$$\psi ::= \lceil \beta \rceil \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K \rangle_{\mathbb{V}}\, \psi \qquad (v \in V).$$

There is no negation on the graded sort, since $\mathbb{C}$ has no canonical one, and $\oplus$ is where interference enters the logic: a disjunction may be false with both disjuncts true.

By Definition 2.6 the clause is evaluated once per candidate rather than once per communication, and $\llbracket \psi_r \rrbracket(\sigma) \in V$ is its value at σ.

Reading a bound name in the clause does not consume it: guard evaluation is pure, which the existing condition evaluator already enforces by construction. This exemption is used throughout §11 and is what allows a clause to depend on both the datum received and the assignment chosen.

6 The quantum namespace

Existing quantum process calculi separate classical from quantum data and use a type system to guarantee that each qubit is owned by a unique process [8, 9]. The rho calculus already has the distinction, in its introduction forms.

Definition 6.1 (The two namespaces)**.** A name is *quantum* if it is bound by `new`, or received on a quantum channel. Every other name — in particular every quotation $@P$ — is *classical*. Quantum names travel only on quantum channels. A clause may take values outside $\mathbb{B}$ only at a receipt whose channels are quantum.

Sorting is by binding structure rather than by syntactic form, because $@(*x)$ is a quotation denoting the `new`-bound name x; reflection makes the surface form unreliable and the binder does not.

Proposition 6.2 (Ownership from unforgeability)**.** *A* `new`*-bound name cannot be denoted by any process that was not given it, whereas $@P$ can be constructed by anyone who can write P. Hence a datum on a quantum channel has at most one prospective consumer per firing, and no process can acquire a reference to a quantum channel except by receiving one.*

Two consequences, both of which the previous version had to argue for.

Remark 6.3 (Contention is not ownership). Unique ownership does not forbid several receipts listening on one quantum channel. A race has exactly one winner per datum, so the race is a competition *over* ownership and not a violation of it. This is the point on which the present design departs from the quantum process calculi: they resolve process non-determinism by scheduling and take probability only from measurement, so the branching of a race is never coherent. Here it is the only place coherence comes from.

Remark 6.4 (The herald is classical by construction). The herald channel of §13 is $h(r) := @r$, the quotation of the site. By Definition 6.1 it is classical. So the boundary between the coherent and the classical parts of a run is drawn by the calculus, and reading a herald is a classical receive — which is where the instrument acts.

Freshness in the classical namespace is not lost: quoting a process that mentions a `new` name, as in $@(c!(*x))$, is fresh and classical.

7 The denotation is register-indexed

A process term cannot be the carrier of a quantum state, for a reason that shows up as soon as one asks how two of them combine: there is no operation taking two quantum states to a quantum state. What there is, is a tensor product of states on *disjoint* systems. So the denotation must be indexed.

Definition 7.1 (Register and state space)**.** The *register* of P, written $\mathrm{reg}(P)$, is the set of quantum names at which P can produce or consume. A name occurring only as inert payload is not in the register. For a register S,

$$H_S := \bigotimes_{x \in S} \mathcal{F}(x),$$

where $\mathcal{F}(x)$ is the Fock space graded by the occupancy of x: the number of messages resting on that channel. $[\![P]\!] \in H_{\mathrm{reg}(P)}$.

Proposition 7.2 (Bosonic statistics are forced by congruence)**.** *Parallel composition is associative and commutative, so a configuration records how many messages sit on a channel and not which is which. Hence the n-occupancy sector of $\mathcal{F}(x)$ is the n-th* symmetric *power, and the amplitude of a coinciding outcome of an m-fold contention is a permanent rather than a determinant.*

This settles, upstream of any choice of normalisation, a question the previous version treated as a fork: whether an m-fold degeneracy carries weight m or m^2 is not a matter of taste but a consequence of the structural rules. A calculus whose parallel composition were not commutative would give a different answer, and there is no such calculus here.

Proposition 7.3 (Composition, and when it forgets)**.** *There is a map $H_S \otimes H_T \to H_{S \cup T}$ adding occupancies, defined for all S, T. It is injective if and only if $S \cap T = \emptyset$. On overlap it identifies $|2\rangle \otimes |0\rangle$, $|1\rangle \otimes |1\rangle$ and $|0\rangle \otimes |2\rangle$, forgetting which side supplied which quanta.*

The forgetting is not a defect. It is indistinguishability, and it is where the permanent of Theorem 9.2 comes from. What genuinely fails on overlapping registers is something else:

Remark 7.4 (The interaction term)*.* $[\![P \mid Q]\!]$ is not determined by $[\![P]\!]$ and $[\![Q]\!]$ when the registers overlap, because $P \mid Q$ has candidates neither factor has. The failure of compositionality is coextensive with the presence of a race (§8), and is the same phenomenon as an interaction term in a joint Hamiltonian. Non-compositionality is also the expected condition in this area rather than a peculiarity of the present design; see [34] and §14.

Remark 7.5 (Separation)*.* A partial tensor defined on disjoint registers is a separation algebra, and the separating conjunction the generated logic of [24] already carries is that tensor. The structure was visible on the logic side before it was visible on the semantics side.

Allocation and discard are now visible as the two directions of one thing. `new` grows the register, adjoining a factor in its vacuum, which is an isometry. Discard shrinks it, which is a partial trace. §13 dilates the second; §10 identifies when the first suffices.

8 Contention-set completion

The one-step operator sums over receipt sites as well as over candidates:

$$T(P) \;=\; \sum_{r} \;\; \sum_{\sigma \in \mathrm{Cand}(P,r)} [\![\psi_r]\!](\sigma) \;\cdot\; \mathrm{out}(r, \sigma). \tag{2}$$

A condition on one site therefore cannot make T norm-preserving. Worse, the outer sum conflates two different things: candidates at different sites may be alternatives, or they may be concurrent, and only the first should be summed.

Definition 8.1 (Conflict, components, matchings)**.** Two candidates *conflict* when they share a datum or share a receipt. A *contention component* at P is a connected component of that relation. A *step* of a component is one of its *maximal matchings*: a maximal set of pairwise non-conflicting candidates.

The terminology follows event structures, where conflict names events that cannot both occur [32]. It is the phenomenon the Born rule is here to decide, and not the failure mode; the failure modes of §10 are called *collisions*.

Proposition 8.2 (Completion)**.** *Let $C_1, \ldots, C_k$ be the contention components at P. Then T factorises as $\bigotimes_i T_{C_i}$, and T preserves norm at P if and only if for each component*

$$\sum_{o \in \mathrm{out}(C_i)} \Big| \sum_{\substack{m \text{ a maximal matching of } C_i \\ \mathrm{out}(m)=o}} w(m) \Big|^2 \;=\; 1.$$

Three things in that statement do work. Components *tensor*: candidates in different components are concurrent and both fire, so treating them as alternatives double-counts. The alternatives within a component are its maximal *matchings*, not its individual candidates — this is the RSpace zip, the bag of data and the bag of patterns admissibly pairable. And the norm is taken *after* summing over the fibre of out, since matchings with coinciding outcomes interfere before they are measured.

Every familiar case is an instance. (1) is one receipt with two data. A single receipt with m patterns and m data has $m!$ matchings, which is Theorem 9.2. Two receipts competing for one datum is the partial-matching case, and its completion condition is literally (1)'s. The distributed join of §4 is two receipts and two data with a one-element fibre.

Measured, over five configurations and four candidate conventions:

configuration	site	candidate	matching	**fibre**
the race equation (1)	1	1	1	**1**
two receipts, one datum	2	1	1	**1**
the distributed join	8	4	2	**1**
the two-fold join	2	2	2	**1**
two independent races	2	1	1	**1**

Table 1: Squared norm of the image under four normalisation conventions; 1 is preservation. Only the fibre convention survives. Normalising over candidates gets the right number on independent races for the wrong reason: it treats concurrency as choice.

Remark 8.3 (Maximal progress on quantum channels). If a component may decline to fire, then "c_1 then c_2" and "both at once" are distinct histories reaching one configuration, and the interleaving inflation Proposition 8.4 exists to remove returns by another door. We therefore take maximal progress *on quantum channels*, leaving classical channels to interleave as usual. Definition 6.1 is what makes that expressible. It has a defensible reading: under a Born-rule resolver an enabled race is resolved, and declining is not among the options.

Proposition 8.4 (Causal histories, not interleavings)**.** *Independent redexes must be counted once. Summing over interleavings instead inflates the squared norm of a layer of n independent sites by $n!$; measured, the inflation is exactly* 2*,* 6 *and* 24 *for $n = 2, 3, 4$ against a causal squared norm of* 1.000000*. Under Definition 8.1 and Remark 8.3 the quotient is not imposed on histories: concurrent firings are bundled into one matching and independent components tensor, so the inflation cannot arise.*

Remark 8.5 (Two honest costs). Completion is *nonlocal*: a clause can be normalised only in the light of what else is contending, so isometry is a property of a configuration and not of a clause. And the sum over maximal matchings is permanent-shaped, hence $\#P$ — which is the physics [2] but bites an implementation, so what a compiler checks must be a syntactic sufficient condition such as a bound on component size.

Conjecture 8.6 (Global norm preservation)**.** *Proposition 8.2 holds at a configuration. That the induced operator preserves norm along arbitrary concurrent execution, with components merging and splitting as data are produced and consumed, is conjectured and not proved.*

9 The join is the recombiner

Interference needs reconvergence, and reconvergence needs the outcome to forget which datum went where. No new term former is required.

Definition 9.1 (Symmetric continuation)**.** P is *symmetric* in $y_1, \ldots, y_m$ when

$$P\{@Q_{\sigma(1)}/y_1, \ldots\} \;\equiv\; P\{@Q_{\tau(1)}/y_1, \ldots\}$$

for all bijections σ, τ and all payloads.

Symmetry is cheap to arrange, because parallel composition is commutative: $z!(*y_1) \mid z!(*y_2)$ is symmetric and $z_1!(*y_1) \mid z_2!(*y_2)$ is not.

Theorem 9.2 (Scattering)**.** *Let C be a contention component with m patterns and m data, and let A_{ij} be the clause value on the matching assigning datum j to pattern i. If the continuations are symmetric then all $m!$ matchings coincide and the outcome carries* $\mathrm{Per}(A)$. *If a clause carries the sign of the permutation, the outcome carries* $\mathrm{Det}(A)$ *instead.*

Corollary 9.3 (Which-path, syntactically)**.** *No clause produces interference at a component whose continuations distinguish their bound names. Whether a race interferes is decided by the continuation, not by the weights.*

The patterns of C need not belong to one receipt. Theorem 9.2 covers the distributed join of §4, which is why the counterexample there is an instance rather than a refutation.

9.1 The minimal interferometer

Take $m = 2$ and a clause returning a on the identity assignment and b on the transposition. With a symmetric continuation there is one outcome with amplitude $a+b$; with a distinguishing continuation there are two, with $|a|^2$ and $|b|^2$. For $a = b = 1/\sqrt{2}$ and relative phase 0, π, $\pi/2$ the coherent weights are 2.000000, 0.000000, 1.000000 against an incoherent 1.000000 throughout. Three lines of rholang, and it is Hong–Ou–Mandel [14]: z is one detector, z_1 and z_2 are two.

10 When is $[\![P]\!]$ a quantum state?

Proposition 8.2 makes each column of T a unit vector. That is half of $T^{\dagger}T = \mathbb{1}$. The other half is that distinct columns be orthogonal, and it fails for reasons that have nothing to do with weights.

Definition 10.1 (Collision)**.** Two configurations occurring in one reachable superposition *collide* when their images share support, so that orthogonality becomes a constraint linking the weights of different configurations. The collision is *severe* when one image is contained in the other, since orthogonality then requires forbidding a transition outright.

Proposition 10.2 (Absorption)**.** *A configuration with no enabled component is a fixed point of T under the fictitious identity jump. If another configuration in the same superposition reduces to it, the two columns are nested and no choice of weights makes them orthogonal. Termination is not a unitary phenomenon.*

Definition 10.3 (Halt tagging)**.** A configuration with no enabled component does not idle; it becomes *halted*, tagged with the step at which it halted, and is thereafter frozen.

The tag records *when*, never *which*: branches halting together share it, so it is a coarsening of history in the same sense as the herald of §13, and by Corollary 9.3 it writes no which-path record.

Definition 10.4 (Spectator)**.** A token not consumed by a step is a *spectator*. A spectator resting on a channel that a firing emits to is indistinguishable from the emitted quantum, by Proposition 7.2.

Definition 10.5 (Staging)**.** A term is *staged* when every receipt emits only to quantum channels that no other receipt emits to and that no spectator occupies — fresh wires per stage, which is what a circuit is.

Measured, over six systems, at one to three tokens and depth three, with collisions counted over all reachable layers:

	plain		halt tagging		tagging + staging	
system	all	severe	all	severe	all	severe
cascade	47	**20**	29	**2**	132	**0**
join	16	**12**	4	**0**	6	**0**
beamsplitter, spectator	6	0	6	0	8	**0**
dual rail, two stages	**0**	**0**	**0**	**0**	**0**	**0**

Table 2: Collisions found by exhaustive search. Of the 32 severe pairs in the plain column, 30 have a normal form on one side; of the 39 non-severe pairs, all 39 involve a spectator. Dual rail with staging has none of either, at every token count and depth searched.

Proposition 10.6 (The quantum-safe fragment)**.** *Let a term have a fixed register, conserve tokens, discard nothing, be halt tagged and staged, and use the dual-rail encoding of Definition 11.3. Then in every configuration searched the columns of T have pairwise disjoint support, so orthogonality is automatic and unitarity reduces to Proposition 8.2, which is local.*

This is stated as a proposition about the search, not as a theorem. What is established is the diagnosis: the two ways a term leaves the fragment are absorption and spectators, and both are visible in the syntax. What is not established is that no third way exists.

Remark 10.7 (Why the previous version saw no symptom). Circuit-shaped terms on fresh channels have one receipt per contention component, so the per-site condition of [13] coincides there with Proposition 8.2, and they are staged, so they collide nowhere. The construction was right; the semantics under it was not. A worked example drawn from the circuit model cannot exercise this part of the theory.

11 Gadgets, and why linearity costs nothing

Definition 11.1 (Linear binding)**.** At a receipt on quantum channels, every bound name is used exactly once in the continuation. Occurrences in the clause do not count, guard evaluation being pure.

Linearity is what makes a step reversible in the bookkeeping sense: weakening is the partial trace and contraction is the copy, and Hilbert space is monoidal rather than cartesian, so a semantics with free discard of amplitude-bearing data cannot be unitary. The previous version declined this restriction on the ground that it costs the gate construction. It does, under the encoding that version used.

Proposition 11.2 (Weakening-free steps are block-diagonal)**.** *Suppose the payload multiset of the outcome is the consumed multiset together with the continuation's literals, and suppose distinct input basis states present distinct payload multisets. Then no outcome lies in the image of two distinct inputs, the one-step operator is block-diagonal in the input basis, and no site realises a basis-changing map.*

The second hypothesis is about the *encoding.* A qubit encoded as which datum sits on a channel satisfies it. A qubit encoded as which channel holds a conserved token does not: every basis state presents the same multiset, differing only in assignment.

Definition 11.3 (Dual-rail encoding)**.** A qubit is a pair of quantum channels, each carrying exactly one of two designated tokens one and zero. The basis states are $q_0!(\mathsf{one}) \mid q_1!(\mathsf{zero})$ and $q_0!(\mathsf{zero}) \mid q_1!(\mathsf{one})$.

Lemma 11.4 (Arbitrary one-qubit gate, linearly)**.** *Let w be a fresh quantum channel carrying the two output rail names, and consider*

$$\textit{for}(\, y \multimapinv q_a \;\&\; z \multimapinv q_b \;\&\; u_1 \multimapinv w \;\&\; u_2 \multimapinv w \textit{ where } \psi \,) \;\{\; u_1!(*y) \mid u_2!(*z) \;\}.$$

Every bound name is used exactly once and nothing is discarded. There are two candidates, the assignments of the output rails to u_1 and u_2; reflection makes a received name usable as the channel of a send, so the candidate chooses where each token lands. Both inputs reach both output configurations, so the outcomes coincide across inputs, and since ψ may read both y and u_1 without consuming them, every 2×2 matrix over $\mathbb{C}$ is realised.

Computed by enumerating the site's candidates: the Hadamard, X, Z, T and a 50:50 beamsplitter are each realised with worst entry deviation 0, as are 500 pseudorandom unitaries; the realised matrix is unitary to 2.2×10^{-16} or exactly. Under the payload encoding the same linearity gives row supports $\{0, 1\}$ and $\{2, 3\}$ with empty overlap — block-diagonal, as Proposition 11.2 says.

Lemma 11.5 (Arbitrary two-qubit gate, linearly)**.** *Four input rails, four output rails and four ancilla patterns give* 24 *candidates per input and* 16 *reachable input/outcome pairs at uniform multiplicity* 4*, so the coinciding candidates form a permanent-shaped sum rather than an artefact. CNOT is realised exactly, as are* 200 *pseudorandom product unitaries.*

Remark 11.6 (What changed, and what did not). The previous version concluded that a design may have structural unitarity or circuit-shaped expressiveness but not both. That was an artefact of Definition 11.3's predecessor. Proposition 11.2 survives with its hypothesis made explicit, and its role is now to explain why the encoding had to change.

12 A worked example: Shor's algorithm

Every number below is computed by a simulator that realises each gate *through the dual-rail site*, enumerating that site's candidates rather than multiplying gate matrices, and the listings are emitted from the same gate list the simulator runs.

12.1 The gadgets

```
// Every channel below is bound by 'new', hence quantum: unforgeable, and the
// only namespace in which a graded clause may take complex values.  Installer
// channels are quantum too; a persistent receipt consumes each datum once and
// uses its bound names once per firing, so persistence never breaks linearity.
// A qubit is ONE token distributed over TWO rails.

contract gate1( r0In, r1In, r0Out, r1Out, m00, m01, m10, m11 ) = {
  new w in {
      w!(*r0Out) | w!(*r1Out)                 // the two output rail names
    | for( y o- *r0In & z o- *r1In & u1 o- w & u2 o- w
           where [ *y == one and *u1 == *r0Out ] * *m00
```

```
              + [ *y == one and *u1 == *r1Out ] * *m10
              + [ *y == zero and *u1 == *r0Out ] * *m01
              + [ *y == zero and *u1 == *r1Out ] * *m11 )
      { u1!(*y) | u2!(*z) }
  }
}

contract hadamard( r0In, r1In, r0Out, r1Out ) = {
  gate1!( *r0In, *r1In, *r0Out, *r1Out,
           0.7071067811865476,  0.7071067811865476,
           0.7071067811865476, -0.7071067811865476 )
}

// Two-qubit gates join four input rails with four ancilla patterns carrying the
// four output rail names.  The clause reads the tokens and the assignment, so
// it determines both the input and the output basis state; entries off the
// intended support return 0 and those candidates do not fire.

// ‘bit‘ reads a rail token as 0 or 1 and ‘rail‘ writes one; both are ordinary
// rholang and neither touches an amplitude.
contract bit( t ) = { if (*t == one) { 0 } else { 1 } }
contract rail( r0, r1, b ) = {
  if (*b == 0) { *r0!(one) | *r1!(zero) } else { *r0!(zero) | *r1!(one) }
}

contract gate2( a0In, a1In, b0In, b1In, a0Out, a1Out, b0Out, b1Out, m ) = {
  new w in {
      w!(*a0Out) | w!(*a1Out) | w!(*b0Out) | w!(*b1Out)
    | for( ya o- *a0In & za o- *a1In & yb o- *b0In & zb o- *b1In
         & u1 o- w & u2 o- w & u3 o- w & u4 o- w
           where entry( *m, basisIn(*ya, *yb), basisOut(*u1, *u3) ) )
      { u1!(*ya) | u2!(*za) | u3!(*yb) | u4!(*zb) }
  }
}
```

12.2 The instance

$N = 15$, $a = 7$, four counting and four work qubits, hence sixteen rail pairs. Each wire carries a stage index, because gadgets compose in series on fresh channels: `q3_5_0` is the first rail of wire 3 after five gates. The staging of Definition 10.5 is exactly this discipline, so the term is in the fragment of Proposition 10.6 by construction.

```
new q0_0_0, q0_0_1, q1_0_0, q1_0_1, q2_0_0, q2_0_1, q3_0_0, q3_0_1,
    q4_0_0, q4_0_1, q5_0_0, q5_0_1, q6_0_0, q6_0_1, q7_0_0, q7_0_1,
    q0_1_0, q0_1_1, q1_1_0, q1_1_1, q2_1_0, q2_1_1, q3_1_0, q3_1_1,
    q0_2_0, q0_2_1, q1_2_0, q1_2_1, q2_2_0, q2_2_1, q3_2_0, q3_2_1,
    q4_1_0, q4_1_1, q5_1_0, q5_1_1, q6_1_0, q6_1_1, q7_1_0, q7_1_1,
    q0_3_0, q0_3_1, q1_3_0, q1_3_1, q0_4_0, q0_4_1, q2_3_0, q2_3_1,
    q0_5_0, q0_5_1, q3_3_0, q3_3_1, q0_6_0, q0_6_1, q1_4_0, q1_4_1,
    q2_4_0, q2_4_1, q1_5_0, q1_5_1, q3_4_0, q3_4_1, q1_6_0, q1_6_1,
    q2_5_0, q2_5_1, q3_5_0, q3_5_1, q2_6_0, q2_6_1, q3_6_0, q3_6_1,
    q0_7_0, q0_7_1, q3_7_0, q3_7_1, q1_7_0, q1_7_1, q2_7_0, q2_7_1,
    out in {

    // 1. the input register, one token per rail pair
    q0_0_0!(one) | q0_0_1!(zero) |
```

```
q1_0_0!(one) | q1_0_1!(zero) |
q2_0_0!(one) | q2_0_1!(zero) |
q3_0_0!(one) | q3_0_1!(zero) |
q4_0_0!(one) | q4_0_1!(zero) |
q5_0_0!(one) | q5_0_1!(zero) |
q6_0_0!(one) | q6_0_1!(zero) |
q7_0_0!(zero) | q7_0_1!(one) |

hadamard!( *q0_0_0, *q0_0_1, *q0_1_0, *q0_1_1 ) |
hadamard!( *q1_0_0, *q1_0_1, *q1_1_0, *q1_1_1 ) |
hadamard!( *q2_0_0, *q2_0_1, *q2_1_0, *q2_1_1 ) |
hadamard!( *q3_0_0, *q3_0_1, *q3_1_0, *q3_1_1 ) |

// 2. modular exponentiation: one candidate, clause value 1,
//    so amplitudes pass through untouched.  Every rail is
//    consumed and every token re-emitted: no discard.
for( x0 o- *q0_1_0 & x0' o- *q0_1_1
   & x1 o- *q1_1_0 & x1' o- *q1_1_1
   & x2 o- *q2_1_0 & x2' o- *q2_1_1
   & x3 o- *q3_1_0 & x3' o- *q3_1_1
   & y0 o- *q4_0_0 & y0' o- *q4_0_1
   & y1 o- *q5_0_0 & y1' o- *q5_0_1
   & y2 o- *q6_0_0 & y2' o- *q6_0_1
   & y3 o- *q7_0_0 & y3' o- *q7_0_1 ) {
  new r in {
    modexp!( *r, 7,
             8 * bit(*x0) + 4 * bit(*x1) + 2 * bit(*x2) + bit(*x3), 15 ) |
    for( m <- r ) {
      q0_2_0!(*x0) | q0_2_1!(*x0') | q1_2_0!(*x1) | q1_2_1!(*x1') |
      q2_2_0!(*x2) | q2_2_1!(*x2') | q3_2_0!(*x3) | q3_2_1!(*x3') |
      rail( q4_1_0, q4_1_1, (*m / 8) % 2 ) |
      rail( q5_1_0, q5_1_1, (*m / 4) % 2 ) |
      rail( q6_1_0, q6_1_1, (*m / 2) % 2 ) |
      rail( q7_1_0, q7_1_1, *m % 2 ) |
      done!(Nil)
    }
  }
} |

// 3. the transform on the counting register
hadamard!( *q0_2_0, *q0_2_1, *q0_3_0, *q0_3_1 ) |
gate2!( *q1_2_0, *q1_2_1, *q0_3_0, *q0_3_1,
        *q1_3_0, *q1_3_1, *q0_4_0, *q0_4_1, cphase(2*pi/4) ) |
gate2!( *q2_2_0, *q2_2_1, *q0_4_0, *q0_4_1,
        *q2_3_0, *q2_3_1, *q0_5_0, *q0_5_1, cphase(2*pi/8) ) |
gate2!( *q3_2_0, *q3_2_1, *q0_5_0, *q0_5_1,
        *q3_3_0, *q3_3_1, *q0_6_0, *q0_6_1, cphase(2*pi/16) ) |
hadamard!( *q1_3_0, *q1_3_1, *q1_4_0, *q1_4_1 ) |
gate2!( *q2_3_0, *q2_3_1, *q1_4_0, *q1_4_1,
        *q2_4_0, *q2_4_1, *q1_5_0, *q1_5_1, cphase(2*pi/4) ) |
gate2!( *q3_3_0, *q3_3_1, *q1_5_0, *q1_5_1,
        *q3_4_0, *q3_4_1, *q1_6_0, *q1_6_1, cphase(2*pi/8) ) |
hadamard!( *q2_4_0, *q2_4_1, *q2_5_0, *q2_5_1 ) |
gate2!( *q3_4_0, *q3_4_1, *q2_5_0, *q2_5_1,
        *q3_5_0, *q3_5_1, *q2_6_0, *q2_6_1, cphase(2*pi/4) ) |
hadamard!( *q3_5_0, *q3_5_1, *q3_6_0, *q3_6_1 ) |
gate2!( *q0_6_0, *q0_6_1, *q3_6_0, *q3_6_1,
```

```
            *q0_7_0, *q0_7_1, *q3_7_0, *q3_7_1, swapMatrix ) |
    gate2!( *q1_6_0, *q1_6_1, *q2_6_0, *q2_6_1,
            *q1_7_0, *q1_7_1, *q2_7_0, *q2_7_1, swapMatrix ) |

    // 4. read the counting register
    for( c0 o- *q0_7_0 & c1 o- *q1_7_0 & c2 o- *q2_7_0 & c3 o- *q3_7_0 ) {
      out!( 8 * bit(*c0) + 4 * bit(*c1) + 2 * bit(*c2) + bit(*c3) )
    }
}
```

12.3 What it computes

Proposition 12.1 (Deterministic communication is amplitude-transparent). *A component with a single maximal matching and a crisp clause contributes the value* 1. *Amplitudes are unchanged by deterministic reduction, and the length of a derivation does not affect its weight.*

Corollary 12.2. *Modular exponentiation is ordinary rholang, running inside every branch at no cost in the semantics. In the listing above it consumes every rail and re-emits every token, so it discards nothing and stays inside Definition 11.1.*

The run fires 8 one-qubit and 8 two-qubit dual-rail sites. The worst deviation between the matrix a gadget realises and the matrix intended is 0. The counting register reads 0, 4, 8 and 12 at 0.250000000 each and everything else at zero, totalling 1.000000000000; continued fractions give $r = 4$, and $\gcd(7^2 - 1, 15) = 3$, $\gcd(7^2 + 1, 15) = 5$. The worst deviation from the payload-encoded run of [13] is 0: the encoding changed and the algorithm did not.

Twelve of the sixteen reachable outcomes carry amplitude zero. By Definition 3.3 they are the interference deficit, so Shor's algorithm in this semantics *is* a large deficit arranged on purpose.

12.4 What the example shows, and what it does not

It shows that linearity is not the obstacle and that the fragment of Proposition 10.6 is expressive enough for the standard algorithms.

It does not show that graded rho has a quantum idiom of its own: every gate is a hand-built gadget and the encoding is circuit-shaped. By Remark 10.7 a circuit-shaped example also cannot exercise §§8–10, which is precisely why those sections are tested by search rather than by this example. The one primitive the design offers that a circuit does not hand over is Theorem 9.2: an m-fold contention is a BosonSampling instance [2] in a single term former, and since matchings are permutation sets the ambient symmetry group is S_m rather than $\mathbb{Z}_N$.

Finally, writing Shor is not running Shor. Classical evaluation of the denotation is exponential; the value of the term is as a compilation source.

13 Dilation, heralds, and the price of erasure

Nothing so far forbids a clause whose matrix is not an isometry, and renormalising at observation is post-selection.

Proposition 13.1 (Overshoot). *With $\mathbb{V} = \mathbb{C}$ and arbitrary clauses, the graded fragment with the Born reading decides* PostBQP, *hence* PP *[1].*

Definition 13.2 (Contractive clause). A clause is *contractive* at a component when its matrix satisfies $\|A\| \leq 1$. Contractivity is a well-formedness judgement, checked at elaboration; for a clause in the normal form of [24] with constant scalars the matrix is a finite table and the check is by inspection.

This is the one thing the design refuses, and it must be refused rather than repaired: a per-site rescaling is not a global scalar, since branches passing through different numbers of sites acquire different factors.

Definition 13.3 (Defect and heralded step)**.** For a contraction A, put $D_A := \sqrt{\mathbb{1} - A^\dagger A}$ and $D_{A^\dagger} := \sqrt{\mathbb{1} - AA^\dagger}$. The heralded step is

$$V_A : |\gamma\rangle \longmapsto A|\gamma\rangle \otimes |\mathsf{ok}\rangle + D_A|\gamma\rangle \otimes |\mathsf{fail}\rangle,$$

with unitary completion the Julia–Halmos two-defect block $\left(\begin{smallmatrix} A & D_{A^\dagger} \\ D_A & -A^\dagger \end{smallmatrix}\right)$ [31]. The two-defect form is used because a matrix arising from a directed rewrite need not be normal.

Proposition 13.4 (The dilation is faithful)**.** *V_A is an isometry and the block is unitary for every contraction A, and the* ok *block is A unaltered — so on the branch where nothing was lost the semantics is unchanged. Computed over the filter clause* $\mathrm{diag}(1, 10^{-3})$ *and* 200 *pseudorandom contractions, the worst deviation of $U^\dagger U$ from the identity is* 2.220×10^{-15} *and the worst deviation of the* ok *block from A is exactly zero.*

The dilation is indexed by the contention component, not by the site and not by a global step; Proposition 8.4 forbids the last of these.

Definition 13.5 (Heralded reduction)**.** The communication rule is amended so that a firing at component r emits one further message: $h(r)!(\mathsf{ok})$ on the A branch and $h(r)!(\mathsf{fail})$ on the defect branch, where $h(r) := @r$. No term former is added; by Remark 6.4 the herald channel is classical.

Proposition 13.6 (The herald carries the defect bit only)**.** *If the herald ranges over* $\{\mathsf{ok}, \mathsf{fail}\}$ *the outcomes of coinciding matchings remain congruent and Theorem 9.2 is unaffected. If it names the matching, every fibre of* out *becomes a singleton and the coherent sum degenerates to the incoherent one at every component of the program.*

Measured on the interferometer of §9.1: bare weights 2.000000, 0.000000, 1.000000; with an ok herald, identical; with a matching-naming herald, 1.000000 throughout.

Corollary 13.7 (Uniform heralding is free where it should be)**.** *At a component whose clause is an isometry the defect branch carries amplitude zero, so the herald reads* ok *on every matching and is inert. Heralding every component therefore costs nothing exactly where the static check applies, which demotes that check from an admission criterion to a compiler optimisation: it says which heralds may be elided.*

Remark 13.8 (Neither traced nor projected, but addressed). A herald left unread at a deficient component is a record, and leaving it unread is a decoherence event — the correct semantics for a program that discards information, not a malformedness to be refused. Reading it is the retry, and by Remark 6.4 reading it is a classical receive. That is where the instrument acts, and it is the only place it acts.

Proposition 13.9 (Post-selection is not a step)**.** *Under Definitions 13.2 and 13.5 no step loses norm, so the renormalisation at observation has nothing to do, and* $\mathrm{diag}(1, \epsilon)$ *is a heralded step followed by a projection onto* ok*. Projection is not a construct of the semantics; it is available only as a retry loop, whose failures are firings.*

Proposition 13.10 (Expected attempts)**.** *If a component heralds* ok *with probability p and the retry loop restores its inputs before each attempt, the expected number of firings before an* ok *is $1/p$ and the expected charge is κ/p for κ the cost of one firing.*

Applied to a Hadamard layer on n wires followed by $\mathrm{diag}(1, 10^{-3})$ on each: surviving norms 0.250001, 0.125000, 0.062500, 0.031250 and success probabilities 0.062500, 0.015625, 0.003906, 0.000977 for $n = 4, 6, 8, 10$, hence expected attempts 16, 64, 256, 1024 — exactly 2^n. The filter that was free costs 2^n firings.

Conjecture 13.11 (Containment by the ledger)**.** *A graded program funded by a budget polynomial in the size of the term does not decide* PP*, since the post-selection witnessing Proposition 13.1 requires an exponentially small success probability, hence by Proposition 13.10 an exponentially large expected charge.*

Proposition 13.10 assumes independent attempts. An adaptive retry schedule is not geometric, and adaptivity is exactly the resource that makes linear optics universal [15]; the two demands meet somewhere that has not been located.

14 Related work, and where this sits

The natural comparison class is not quantum process calculi but *quantum control.*

CQP [8, 9] adds measurement and unitary transformation to the pi calculus, transmits qubits along channels, and uses a type system to guarantee that each qubit has a unique owner. Its operational semantics combines non-determinism arising exactly as in the pi calculus with the probabilistic results of measurement; the quantum state is a separate component of the configuration and norm preservation is an invariant of that component. Process branching is never coherent. Later work carries distributions over configurations [23], but they remain probability distributions. That architecture does not face the question of this paper, because it declines the premise: here the branching of a race *is* the quantum state.

The literature that does face it is quantum alternation. QML [3] superposes branches under an orthogonality judgement; QGCL [34] does it with an auxiliary system of quantum coins — a control register, which is the branch register of the compilation route — and notes that the resulting superoperator semantics is not compositional. Bădescu and Panangaden [4] keep the control superposed and show that the semantics must then be given by Kraus decompositions rather than superoperators.

Three of their findings bear on this design. Proposition 3.1 is one of them, and it converts §3's negative result from an argument into a citation. Non-compositionality is the expected condition in this area, which is what Remark 7.4 leans on. And their proposition that quantum alternation is not monotone in the Löwner order, hence incompatible with the standard fixed-point semantics for recursion, applies here and is not repaired by anything in this paper; see Obligation 15.3.

Their suggested way out is worth quoting in substance. They propose looking at quantum optics, where a Mach–Zehnder interferometer is a physically real alternation, and they observe that in such a device the system being split is the system the alternate operations act on, with no distinct control qubit. That is the architecture developed here: the race is the split, there is no control register, and Definition 11.3 makes the interferometer literal. Ying's other route to recursion, via second quantisation [33], is also where Proposition 7.2 lands.

The weighted-history pattern itself — multiply along a path, sum across paths — is standard in semiring-weighted computation [7], and stochastic process calculi are long established [22, 29]. The specific claim here is narrower and, we think, new: a semiring-valued expression placed exactly at candidate-match resolution, with rho's joins as explicit recombiners of weighted alternative histories.

15 Open problems

Obligation 15.1 (Global norm preservation)**.** Conjecture 8.6: prove Proposition 8.2 along arbitrary concurrent execution, with components merging and splitting.

Obligation 15.2 (Staging as a theorem)**.** Table 2 is a search. Prove that a halt-tagged, staged, dual-rail term has pairwise disjoint columns, or find the third failure mode.

Obligation 15.3 (Recursion and the Löwner order)**.** Quantum alternation is not monotone in the standard order on completely positive maps [4], so it is incompatible with the usual fixed-point semantics for recursion. Rho's recursion is by reflection, and the budget in the denotation is precisely a stand-in for a fixed point. This is the deepest open problem here and nothing above addresses it.

Obligation 15.4 (The price, made exact)**.** Conjecture 13.11 assumes independent attempts, and the charge for a herald has not been fixed against the cost model. Until it is, "polynomially funded" names an intention.

Obligation 15.5 (Herald garbage)**.** Heralds at deficient components accumulate. Collecting them without reading them — which commits to an outcome — or discarding them — which is the trace this design avoids — is open.

Obligation 15.6 (Categorical status)**.** The register-indexed denotation with its partial tensor is a separation algebra valued in Fock spaces. Making Remark 7.5 precise, and relating it to the generated logic's separating conjunction, is the natural next step.

16 Conclusion

The rho calculus does not say how its races are settled, and quantum mechanics is one of the available answers. Taking that seriously turns out to require four things that were not in the original design.

A term is not a quantum state. The denotation must be indexed by a register of unforgeable names, valued in Fock spaces over those names, with a tensor that is partial and forgets which side supplied which quanta when the registers overlap. The forgetting is indistinguishability; and because parallel composition is associative-commutative, bosonic statistics — and the permanent of the scattering matrix — follow from the structural rules rather than from a choice.

The unit at which amplitudes must be completed is the contention set: connected components of the conflict relation, with maximal matchings as the alternatives and the fibre-summed amplitudes normalised. Components tensor. The race equation we began from is the smallest instance.

Not every term denotes a state, and the ways of failing are syntactic. Absorption — termination, which is not a unitary phenomenon — and spectators — parked quanta indistinguishable from emitted ones — are the two we have found, and halt tagging, staging and a dual-rail encoding remove them in every case we have been able to construct.

And what cannot be excluded can be priced. Contractivity is refused; the residual defect is dilated at the component; the ancilla is neither traced nor projected but written, as a message in the classical namespace, which is where measurement was always going to have to happen. A programmer who wants post-selection may have it, as a retry loop, at 2^n firings.

What remains is not small. Norm preservation is proved at a configuration and conjectured in general; staging is measured and not proved; and quantum alternation's incompatibility with the standard semantics of recursion is a problem this design inherits and does not solve. But the shape of the object is now clear enough to say what would settle it.

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